



Marine Biology

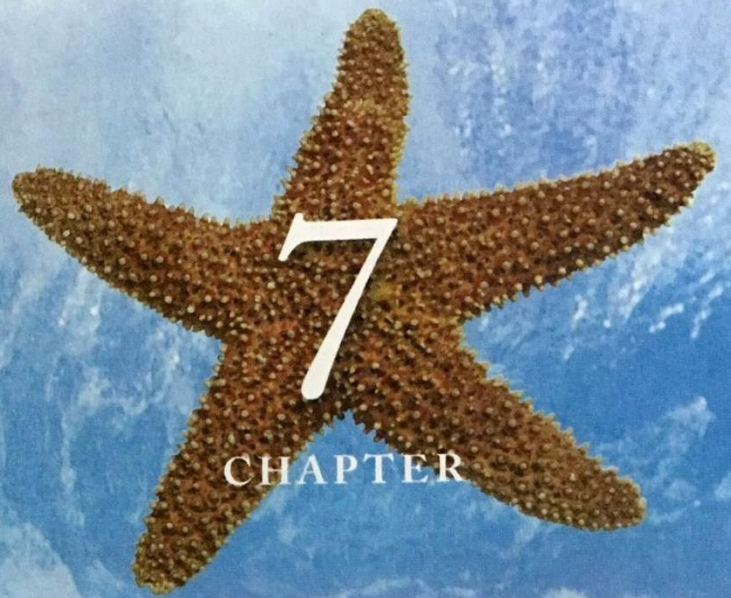
海洋生物学



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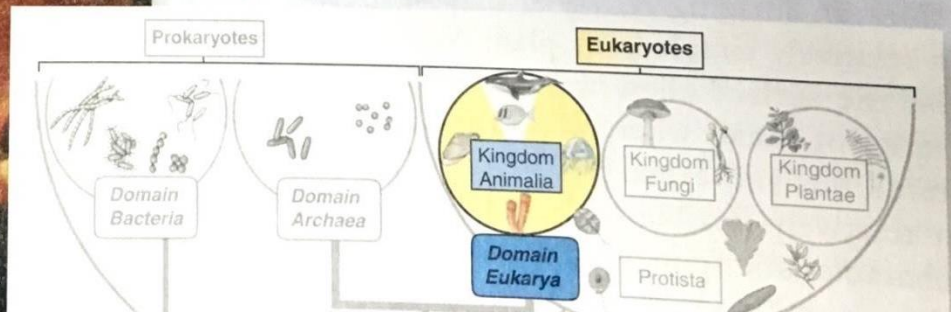
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7

CHAPTER

Marine Animals Without a Backbone



• Major Phyla of Marine Invertebrates

- | | |
|-----------------------|----------------------------|
| 1. Sponges (海绵动物) | 10. Molluscs (软体动物) |
| 2. Cnidarians (腔肠动物) | 11. Bryozoans (苔藓虫) |
| 3. Flatworms (扁形虫) | 12. Phoronids (帚虫) |
| 4. Ribbon worms (纽虫) | 13. Brachiopods (腕足类) |
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| 6. Annelids (环节动物) | 15. Echinoderms (棘皮动物) |
| 7. Sipunculans (星虫动物) | 16. Hemichordates (半索海生动物) |
| 8. Echiurans (螠虫动物) | 17. Tunicates (被囊动物) |
| 9. Arthropods (节肢动物) | 18. Lancelets (文昌鱼) |

Phylum

Arthropods

Subphylum

Protoarthropoda
(原节肢动物)

Euarthropoda
(真节肢动物)

Class

Onychophora
(有爪纲)



- Arachnoida (蛛形纲)
- Insecta (昆虫纲)
- Myriapoda (多足纲)
- Merostomata (肢口纲)
- Crustacea (甲壳纲)



Copepoda
(桡足亚纲)

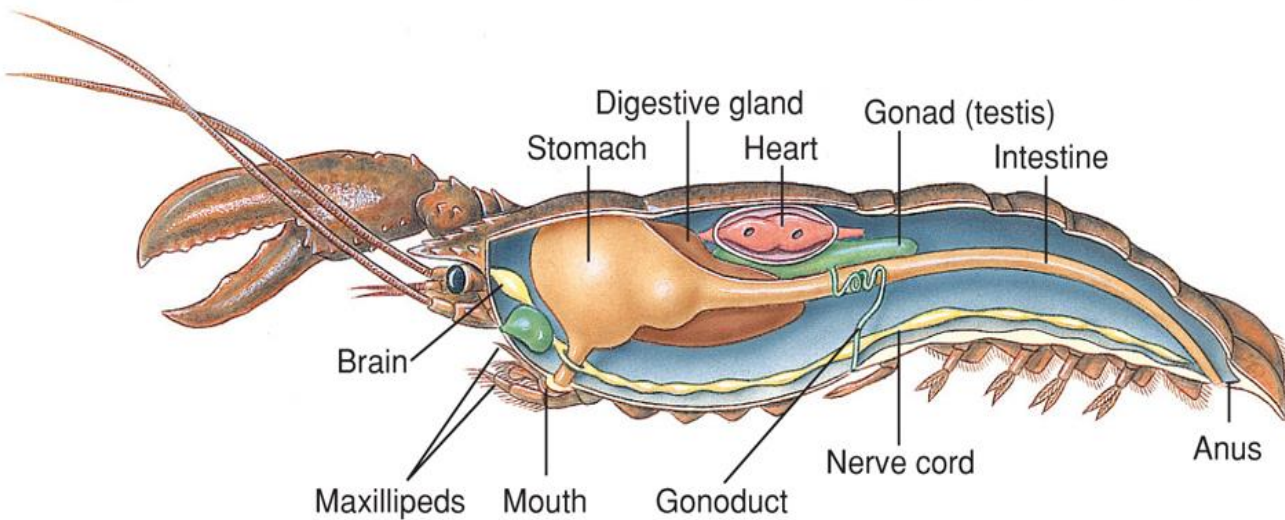
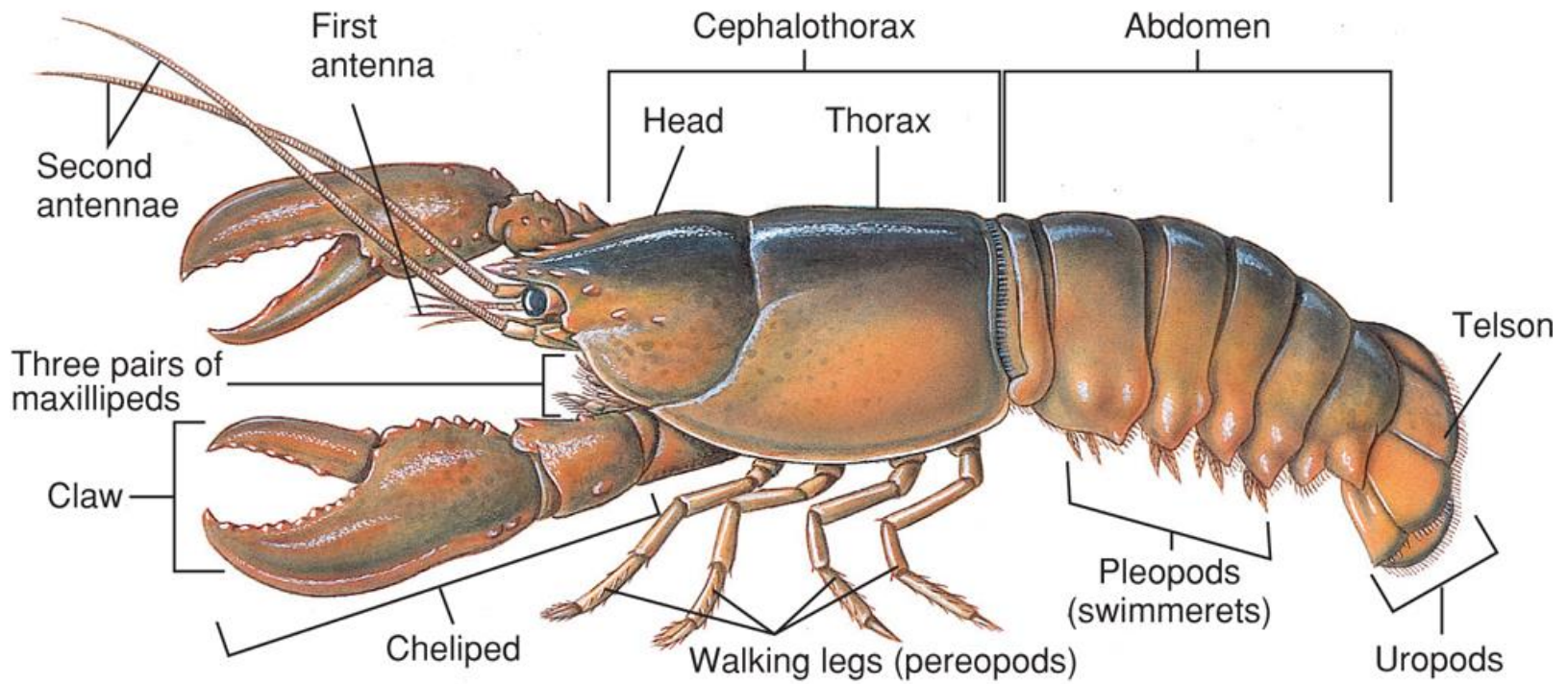
Cirripedia
(蔓足亚纲)

Malacostraca
(软甲亚纲)

Example 1: Shrimps

- Have laterally compressed bodies with distinct and elongated abdomens (we call it 'tail' and like to eat)
- Shrimp are typically scavengers (食腐), specialists in the feeding on bits of detritus on the bottom
- Many colorful shrimps live on the surface of other invertebrates or remove parasites from the skin of fish
- Ghost and mud shrimps burrow in muddy bottoms





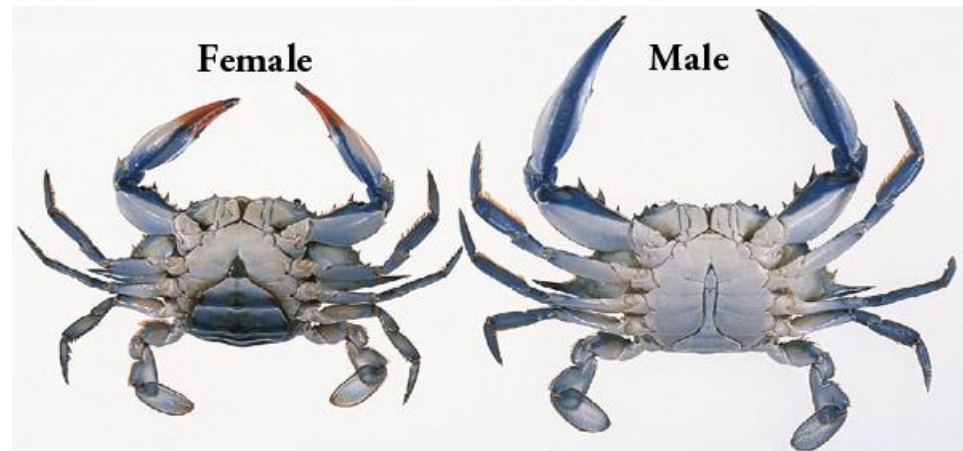
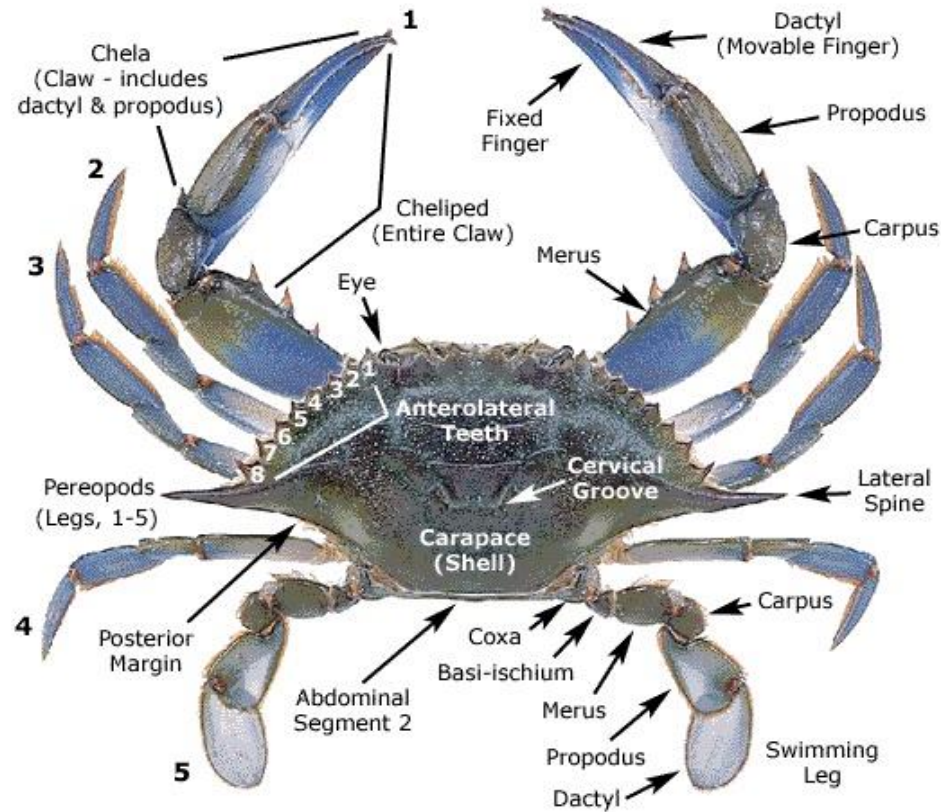
Example 2: Lobsters

Lobsters are mostly nocturnal (夜间活动的) and hide during the day in rock or coral crevices (裂缝), scavengers and predators that crush mollusks and sea urchins



Example 3: Crab

- ❑ The abdomen is small and tucked under the compact and typically broad cephalothorax
- ❑ The abdomen is visible flat
- ❑ V-shaped plate in males, expended and become U-shape for carrying eggs in females
- ❑ Highly mobile and most can easily move sideways when in a hurry



- ❑ Make up **the largest** and most diverse group of decapods, more than 4,500 species
- ❑ Most are scavengers or predators
- ❑ Live along rocky shores or sandy beaches, exposed to air much of the time
- ❑ Land crabs spend most of their lives on land, returning to the sea only to release their eggs





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Example 4: Hermit crab

- ❑ Hermit crab (not true crabs) are scavengers, they hide their long, soft abdomens in empty gastropod shells
- ❑ Some hermit crabs cover their shells with sea anemones or sponges for added protection and camouflage

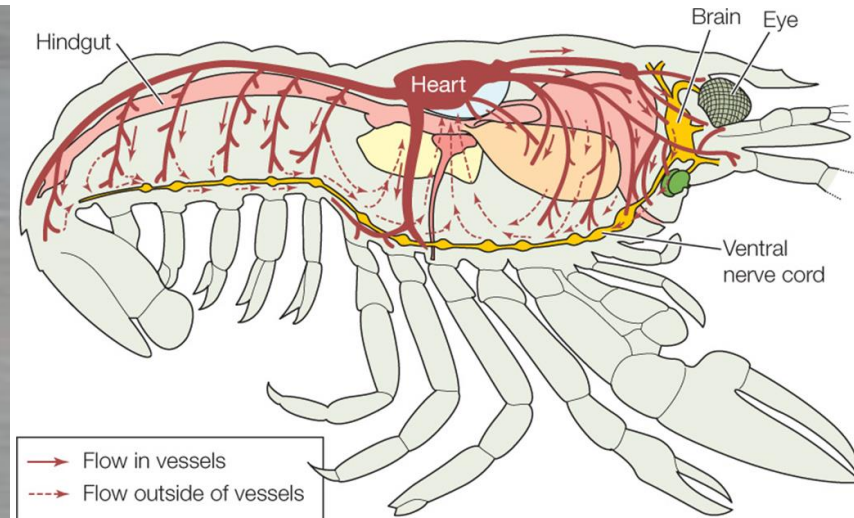


7) Biology of Crustaceans

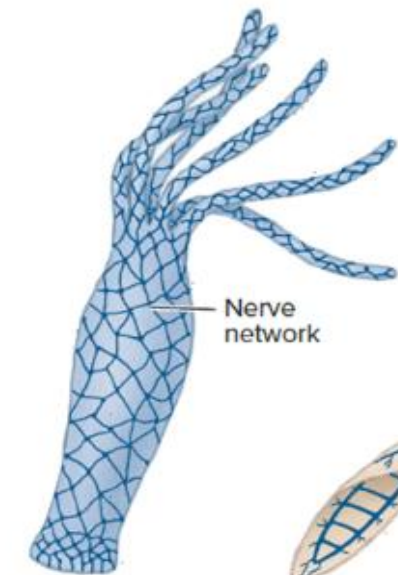
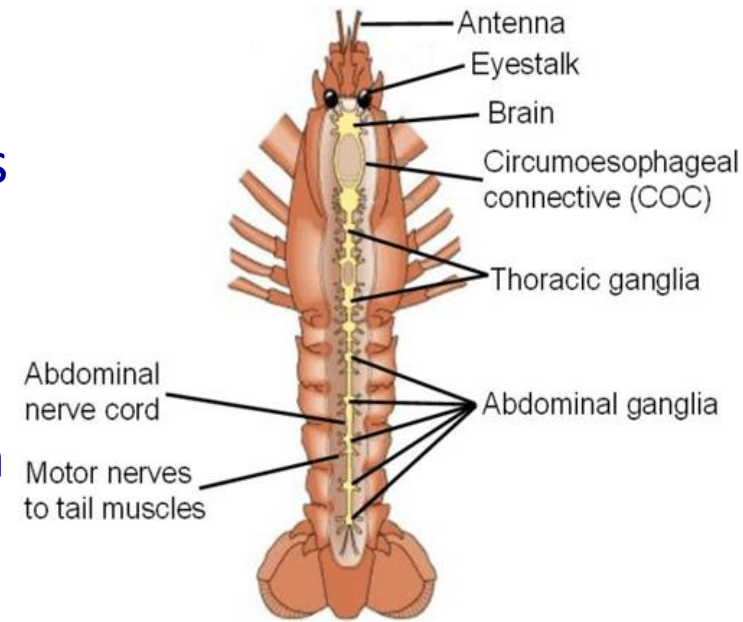
- Feeding and digestion

- ❑ Filter feeder or catching food by stiff, hair-like bristles on appendages;
- ❑ Some appendages are adapted for piercing and sucking in parasitic copepods and isopods
- ❑ Stomach has **chitinous teeth** or **ridges** for grinding; stomach has **two chambers** in decapods
- ❑ Stomach connected to digestive gland which secrete digestive enzymes and absorb nutrients
- ❑ Extracellular digestion, the intestine ends in an anus

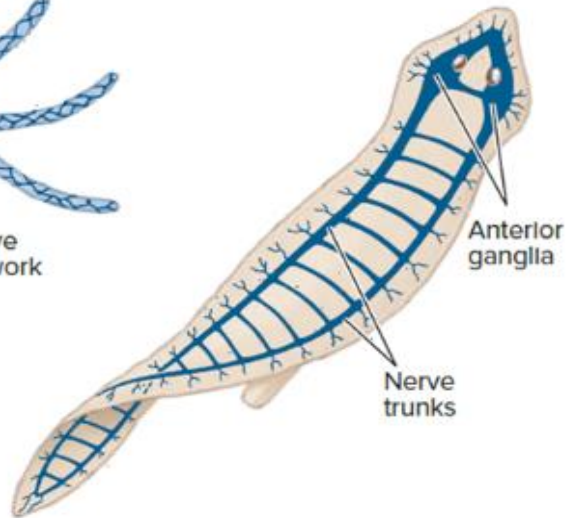
- ❑ Absorbed nutrients are distributed by an **open circulatory system**
- ❑ Gas exchange is typically carried out by gills attached to some of the appendages
- ❑ In most decapods, the gills lie in a chamber under the carapace, where they are constantly bathed by water
- ❑ In land crab, the gill chamber, through moistened, is filled with air and acts almost as our lungs do



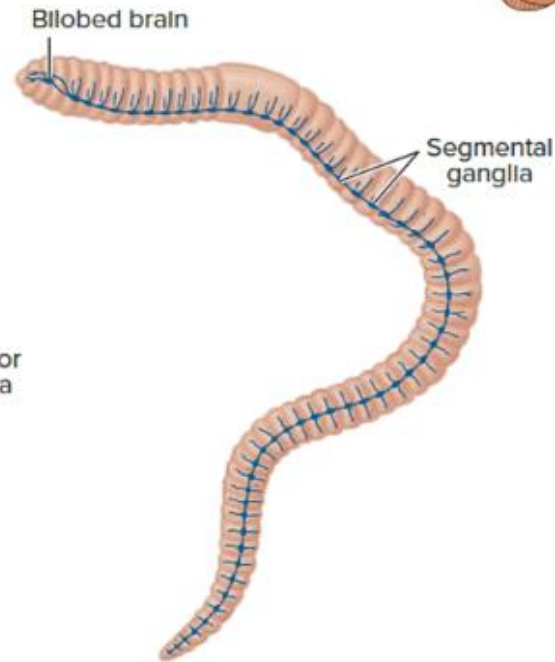
- Nervous system and behavior
 - Nervous system in simple crustaceans is **ladder-like**, but more centralized in decapods
 - Having a small and relative simple **brain** but **well-developed sensory organs**



A Hydra

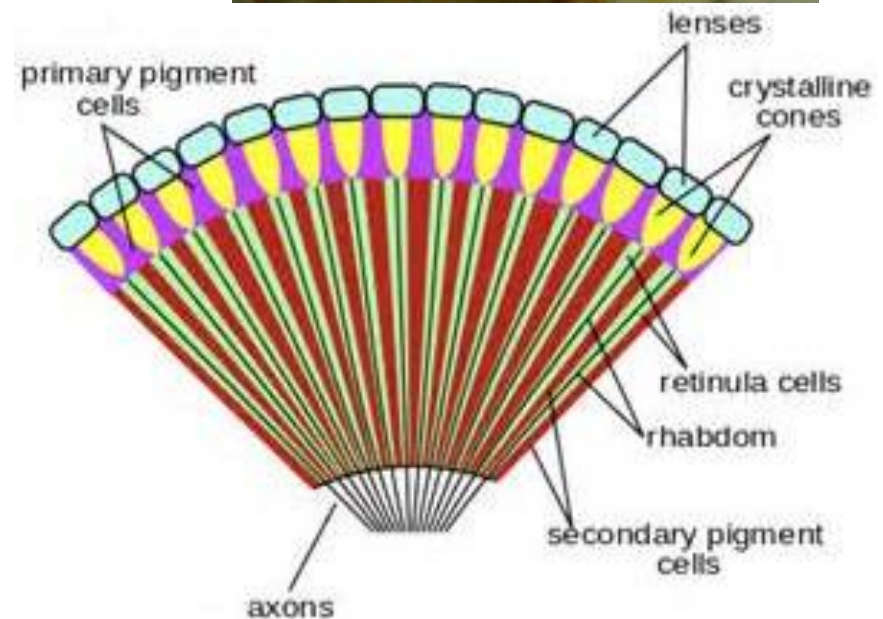
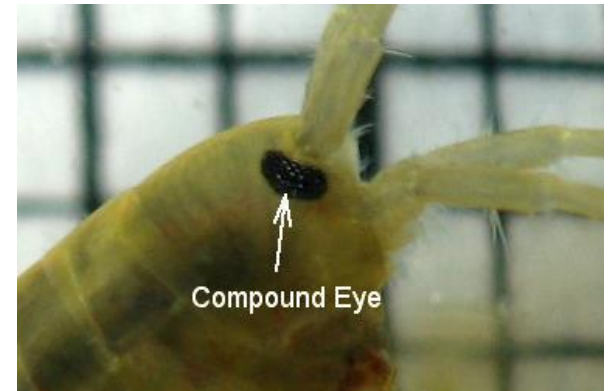
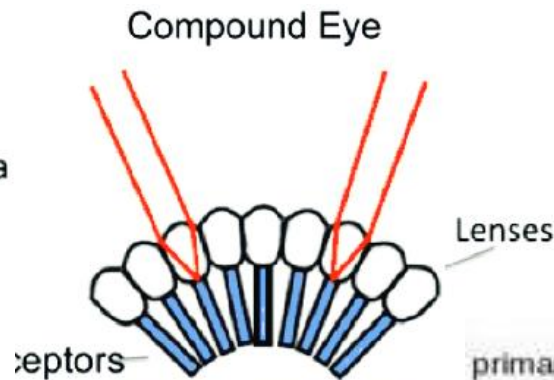
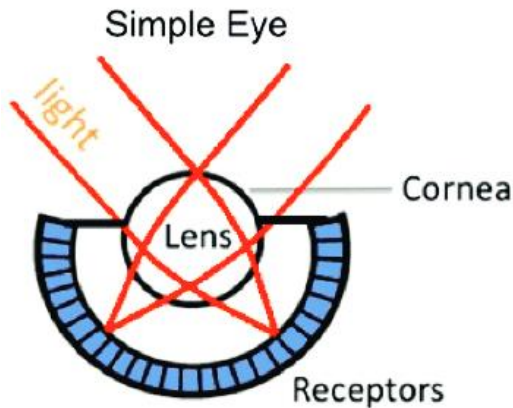


B Planaria

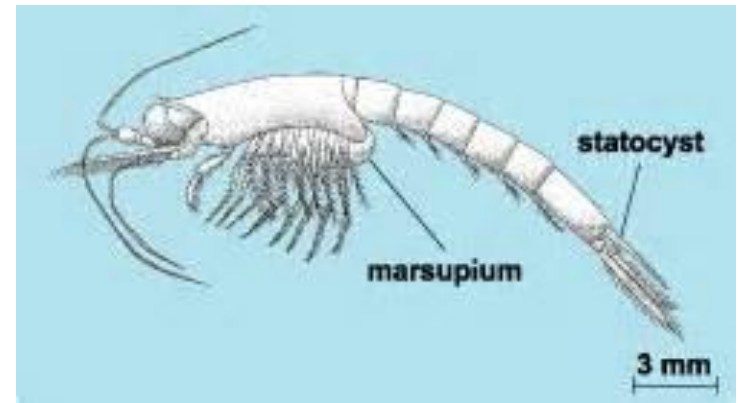
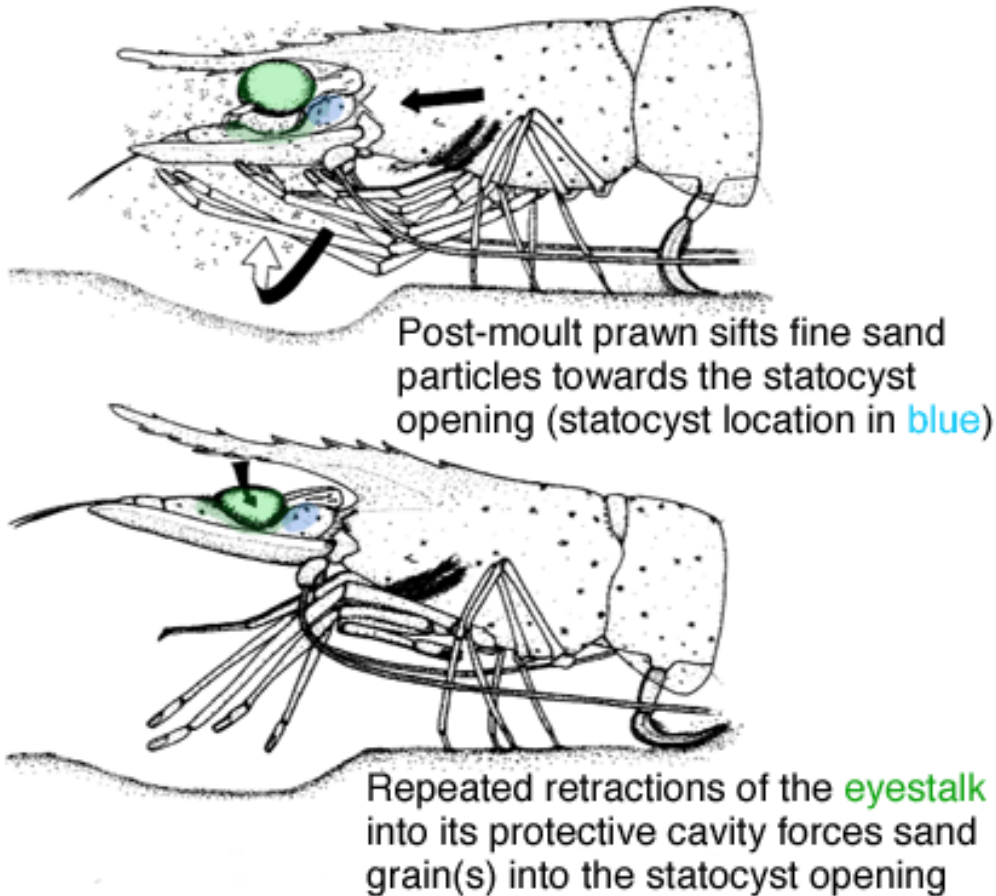


C Earthworm

- Most have **compound eyes**, which consist of a bundle of up to 14,000 light-sensitive units grouped in a mosaic
- In decapods, the compound eyes are at the end of **movable stalks** and can be used as periscopes



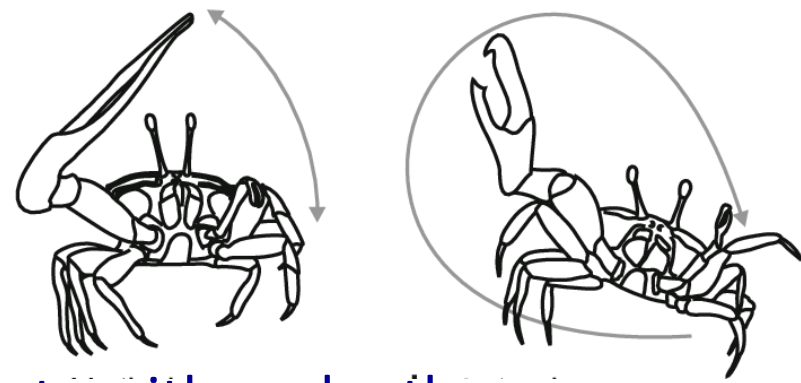
- Have a keen sense of “smell”, which make them sensitive to chemicals in the water
- Have a **pair of statocysts (平衡胞)** for balance



▣ Behaviorally complex invertebrates:

→ Using a variety of signals to communicate with each other

→ Signals involve special body postures or movements of the appendages and antennae, which are often distinctively marked or colored to make the signals more conspicuous (显而易见)



- Reproduction and life history

- ❑ The sexes mature are separate in most crustaceans (October for male and April-May for female)
- ❑ Males use specialized appendages to transfer sperm directly to the female. Even hermaphroditic species transfer sperm between individuals
- ❑ Mating in decapods usually takes place immediately after the female molts, while the exoskeleton is still soft



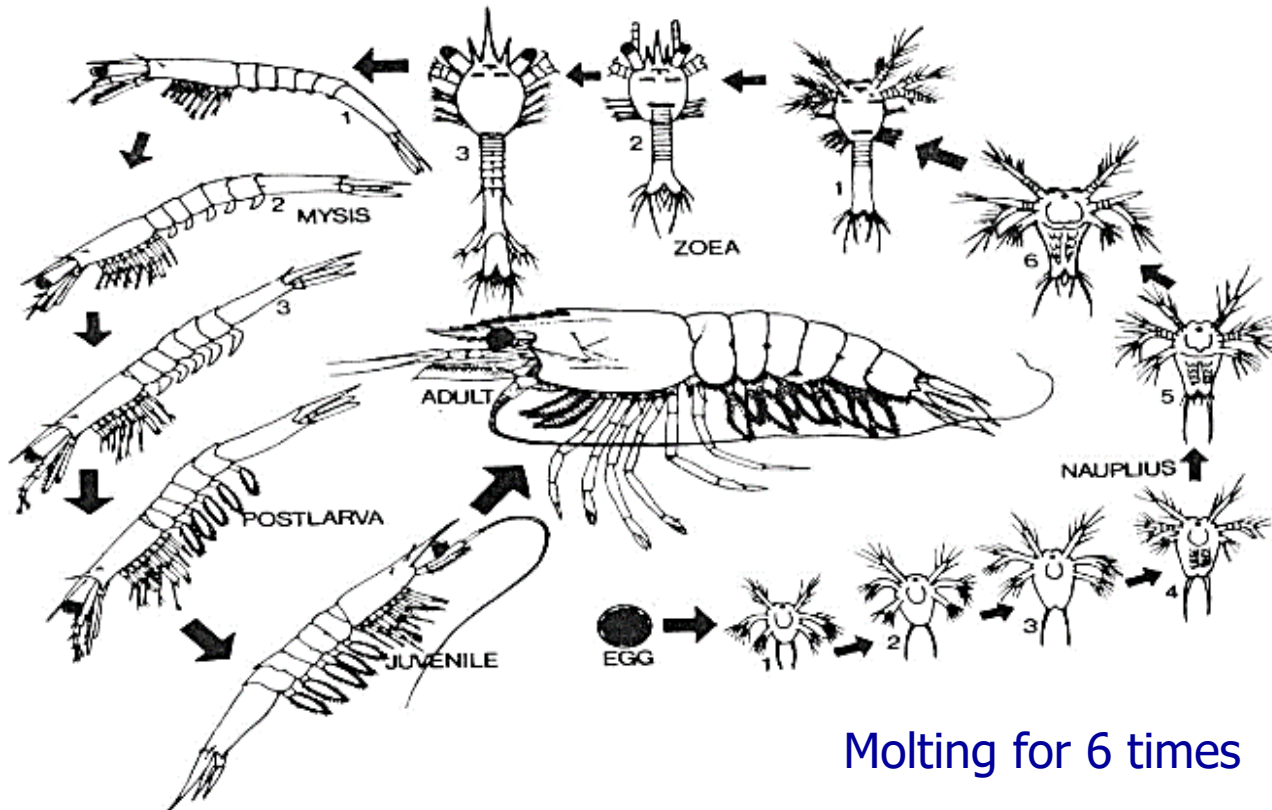
That's not a stalker –
that's my old shell.

- ❑ Females of many species can store sperm for long periods and use it to fertilize separate batches of eggs:
 - In amphipods and isopods eggs are brooded in a chamber formed by lateral extensions of the body
 - In decapods and other groups females carry their eggs using pleopods (游泳足), or swimmerets, specialized appendages beneath the body

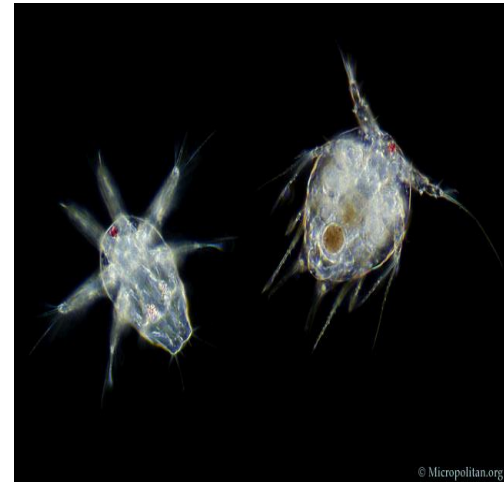


- Most crustaceans have planktonic larvae that look nothing like the adult
- Probably the **most characteristic crustacean larva** is the **nauplius (无节幼体)**, but the type and number of larva stages vary

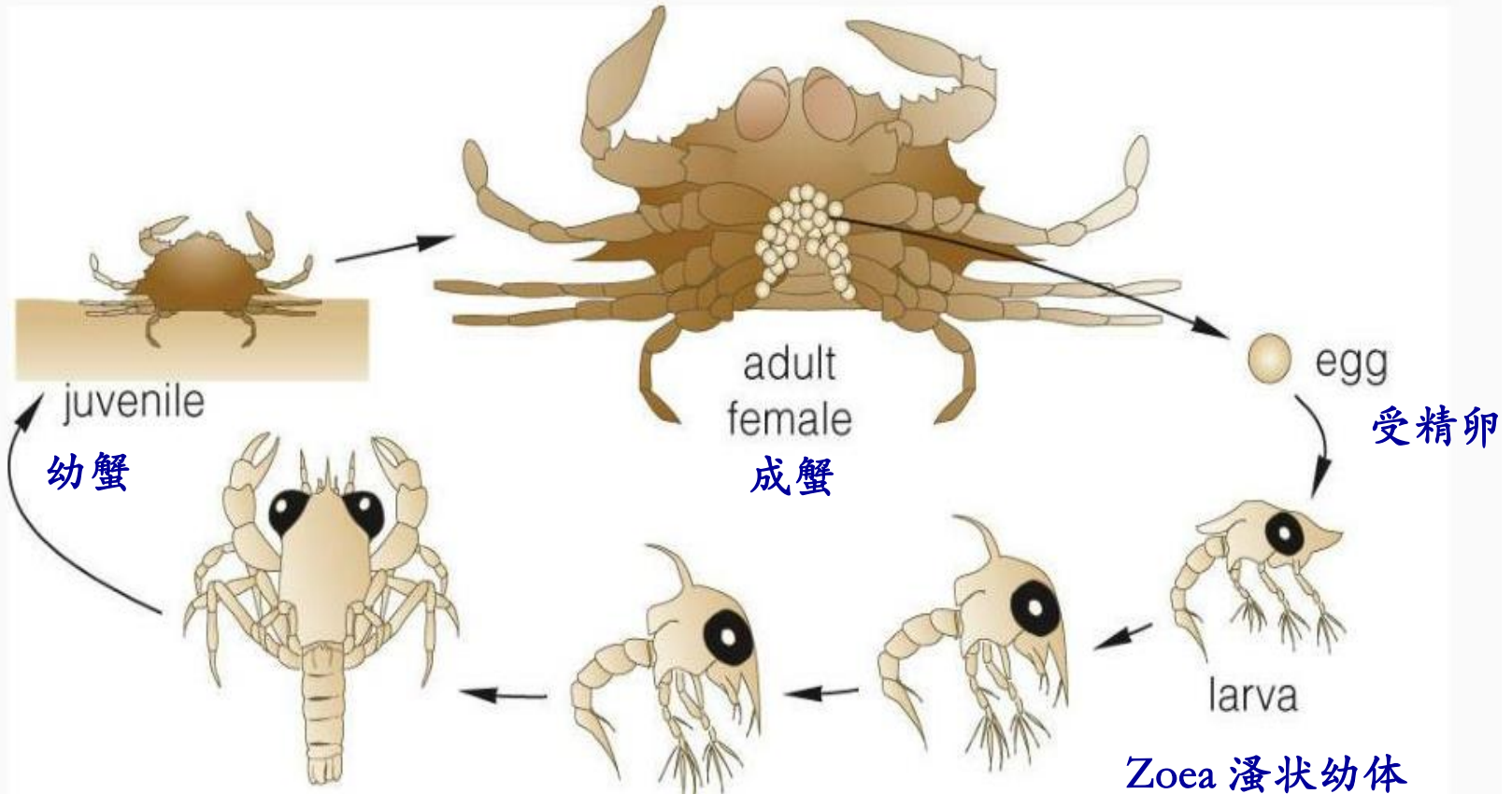
Example: Shrimp's reproduction cycle



Molting for 6 times



Example: Crab's reproduction cycle



Egg
受精卵



Zoea
蚤状幼体



Megalopa
大眼幼体



Juvenile
幼蟹



Adult
成蟹

10.3 Horseshoe crabs (马蹄蟹, 鲎) — Merostomata (肢口纲)

- **5 pairs of legs**, first pair modified in males for reproduction
- Females larger than males
- Mating pairs come onto beaches each spring to breed and lay their eggs in wet sand
- Among the oldest creatures on earth – they have remained virtually unchanged for millions of years
- They live and borrow in soft sediments, normally near shore where they feed on other invertebrates and scavenge

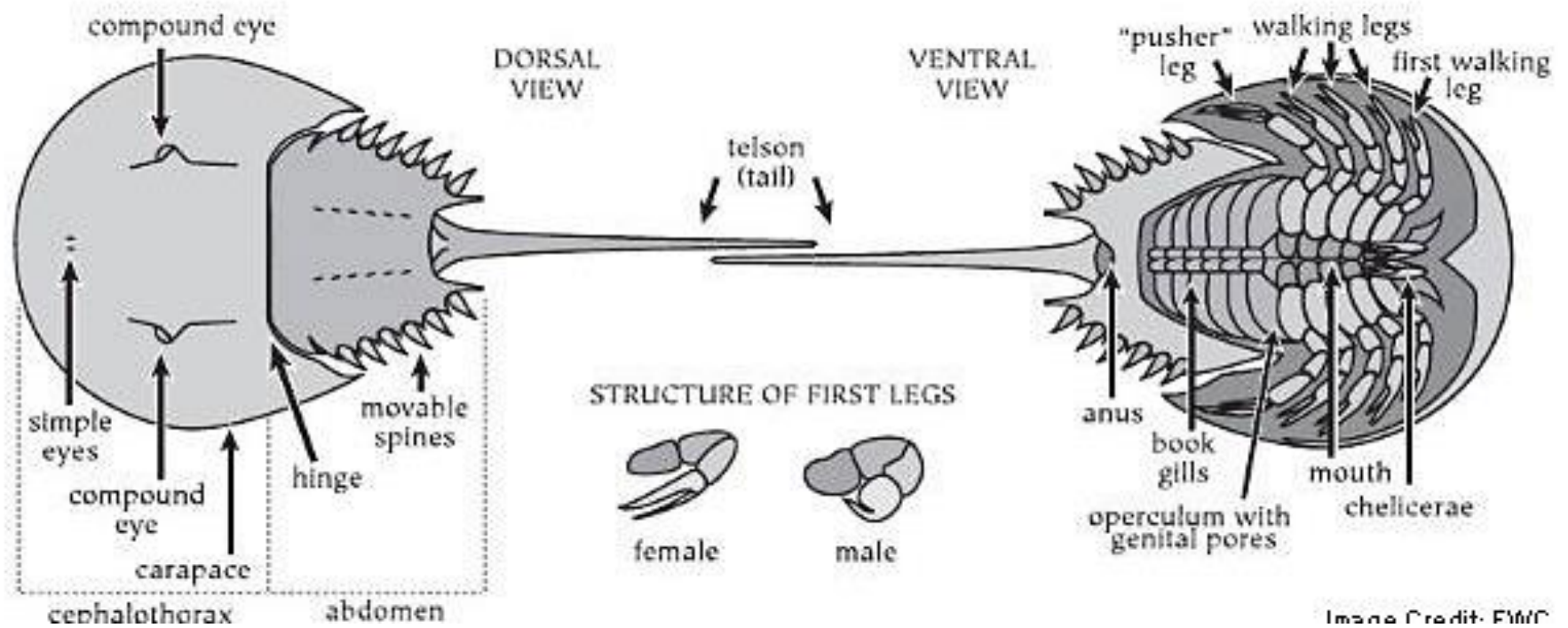


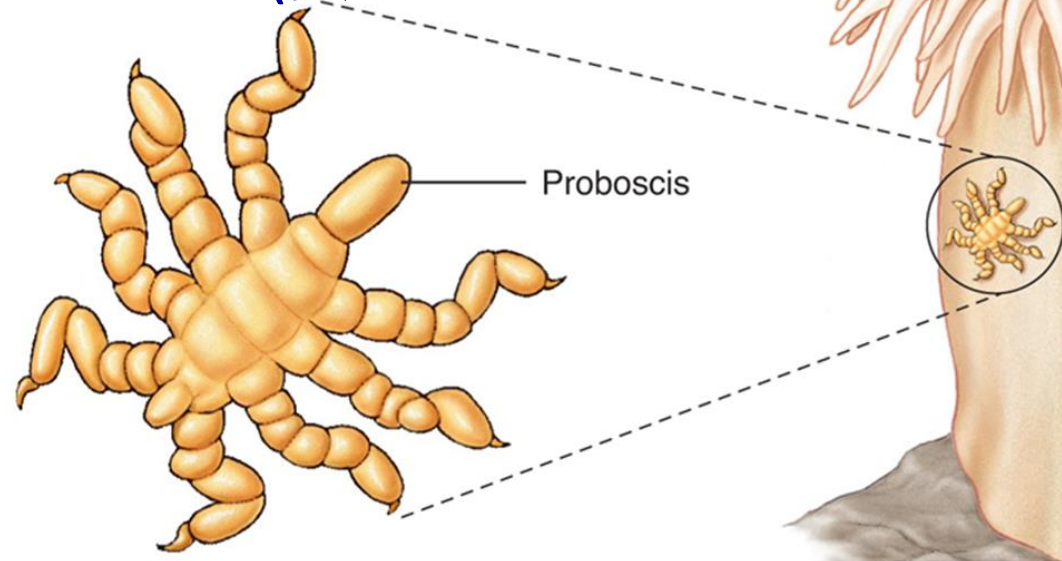
Image Credit: EMMO





10.4 Sea Spiders:

- Four or more pairs of jointed legs, not insects or true spiders
- Possess a mouth and proboscis for feeding
- Mainly feed on sea anemones and hydrozoans (they are voracious (贪婪的) predators!)
- More common in cold waters, but can be found worldwide



10.5 Insects:

- Very few marine insects exist
- Many insects feed in the intertidal zone at low tide, but these are just temporary visitors

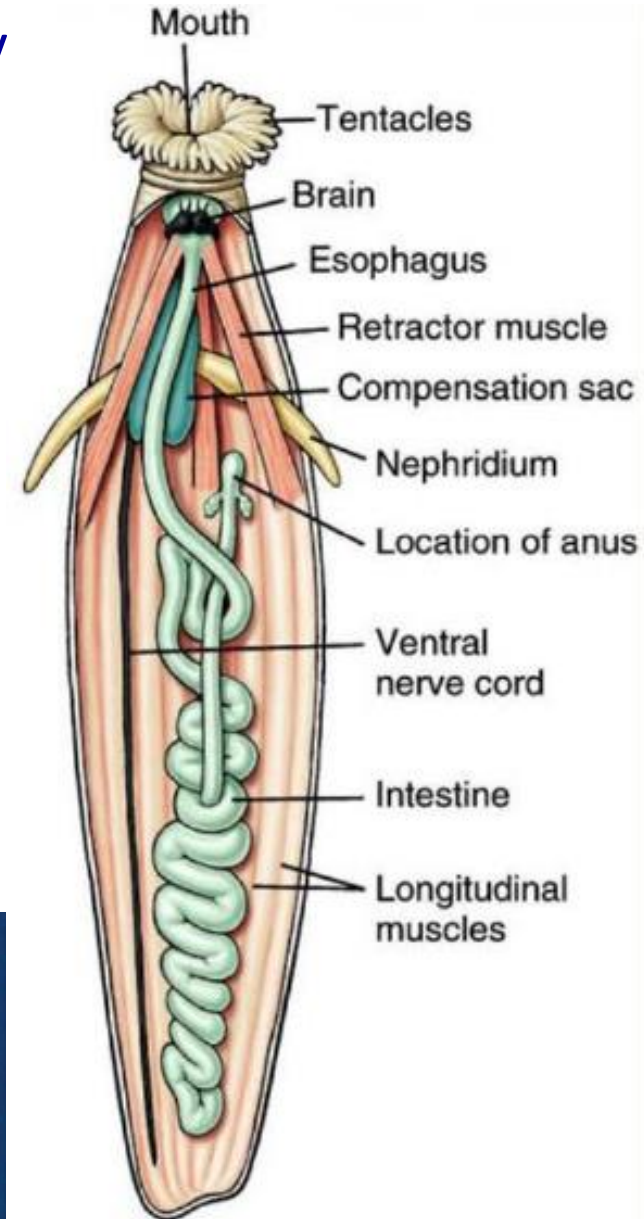


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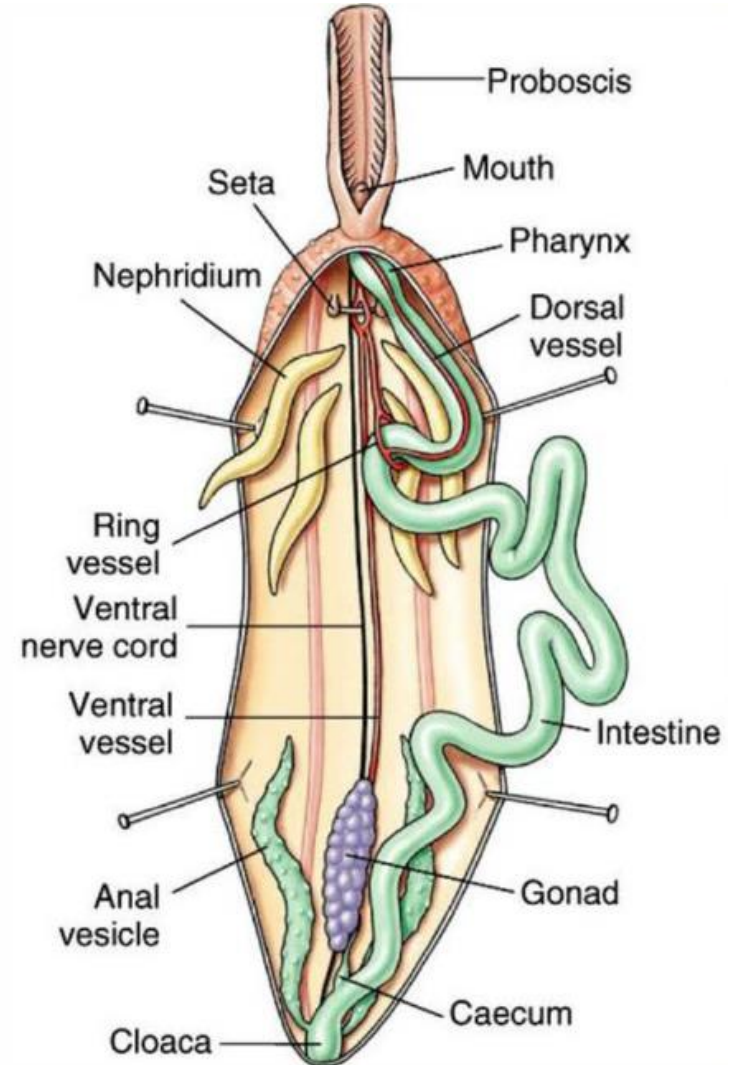
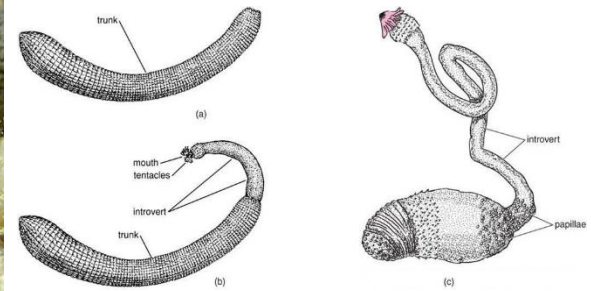
8. Sipunculans (星虫动物) or Peanut Worms

- About 320 species, **all marine**, found mainly in shallow coastal waters, sometimes found in corals or in empty shells of other animals
- Deposit feeders, normally burrow into soft sediments
- Their soft unsegmented bodies possess a retractable multi-lobed or tentacle anterior used for feeding
- **Trochophore**



9. Echiurans (螠虫)

- 135 species
- Non-retractable spoon-shaped or forked proboscis (feeding structure), **trochophore**
- Some taxonomists classify them with annelids
- Deposit feeder, some burrow in soft bottoms, others live in coral



- Homework

